Pure Elegance & Structural Logic

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Part 1 Algebra · Geometry · Combinatorics

Ten short problems at the level of the TMUA and the MAT. Each one has a solution of a few lines once you find the right idea, usually an invariant, a symmetry, a factorisation or a geometric picture. If your solution needs a page of algebra or a coordinate bash, look for another approach. Section 1 has the problems and Section 2 the solutions.
Prerequisites. A-level Mathematics, and complex numbers in exponential form for Problem 9.

1 Problems

Problem 1 Let x, y and z be real numbers such that $x+y+z=0$ and $x^2+y^2+z^2=2$. Find the exact value of

$$x^4+y^4+z^4.$$

Problem 2 Find the minimum value of the following expression for real x.

$$\sqrt{x^2+4}+\sqrt{x^2-10x+29}$$

Problem 3 Find the exact value of the infinite series

$$\sum_{n=1}^{\infty}\frac{n}{n^4+4}.$$

Problem 4 Determine the number of solutions to the equation

$$\sin^{2026}(x)+\cos^{2026}(x)=1$$

in the interval $0\le x\le 2\pi$.

Problem 5 Let $f(x)$ be a continuous function defined for all real x such that

$$f(x)+f(1-x)=x^2-x+\frac{1}{2}.$$

Find the exact value of the definite integral $\int_0^1 f(x)\,dx$.

Problem 6 A whiteboard initially contains the fractions $\frac{1}{2},\frac{1}{3},\frac{1}{4},\ldots,\frac{1}{2026}$. During each turn, a student chooses two numbers on the board, say a and b, erases them, and replaces them with the single number $a+b-ab$. This process is repeated until exactly one number remains on the board. What is this final number?

Problem 7 Find all polynomials $P(x)$ with real coefficients such that for all real x,

$$xP(x-1)=(x-3)P(x).$$

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Problem 8 51 points are placed arbitrarily inside a square of side length 1. Prove that there must exist some set of three points among them that can be completely covered by a closed disc of radius $\frac{1}{7}$.

Problem 9 Find a cubic equation with integer coefficients whose roots are exactly $\cos\left(\frac{2\pi}{7}\right)$, $\cos\left(\frac{4\pi}{7}\right)$ and $\cos\left(\frac{6\pi}{7}\right)$.

Problem 10 Find all real triples (x, y, z) that satisfy the system of equations

$$x + y = z^2, \qquad y + z = x^2, \qquad z + x = y^2.$$

2 Solutions

Problem 1

Since $(x+y+z)^2 = x^2+y^2+z^2+2(xy+yz+zx)$, the given values give $xy+yz+zx = -1$. Next we square the sum of squares,

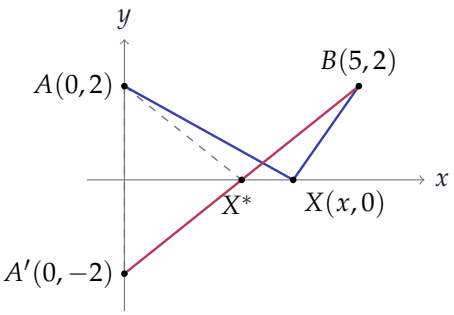
$$(x^2+y^2+z^2)^2 = x^4+y^4+z^4+2(x^2y^2+y^2z^2+z^2x^2),$$

so we need the cross term. Note that $(xy+yz+zx)^2 = x^2y^2+y^2z^2+z^2x^2+2xyz(x+y+z)$, and since $x+y+z=0$ this gives $x^2y^2+y^2z^2+z^2x^2=1$. Substituting back, $2^2 = x^4+y^4+z^4+2\cdot 1$, hence $x^4+y^4+z^4=2$. ■

Problem 2

Rewrite the expression as

$$\sqrt{(x-0)^2+(0-2)^2}+\sqrt{(x-5)^2+(0-2)^2}.$$



This is the sum of the distances from the point $X(x,0)$ on the x-axis to the points $A(0,2)$ and $B(5,2)$. To minimise this sum, reflect A in the x-axis to get $A'(0,-2)$. The shortest path from A' to B is a straight line, which meets the x-axis at the optimal point X^*. By the distance formula, the minimum value is

$$A'B = \sqrt{(5-0)^2+(2-(-2))^2} = \sqrt{25+16} = \sqrt{41}.$$

Problem 3

Factor the denominator with the Sophie Germain identity,

$$n^4+4 = (n^2+2)^2-4n^2 = (n^2-2n+2)(n^2+2n+2) = \big((n-1)^2+1\big)\big((n+1)^2+1\big).$$

Partial fractions now give

$$\frac{n}{n^4+4} = \frac{1}{4}\left(\frac{1}{(n-1)^2+1}-\frac{1}{(n+1)^2+1}\right).$$

Let $u_n = \frac{1}{(n-1)^2+1}$. The sum is $\frac{1}{4}\sum_{n=1}^{\infty}(u_n-u_{n+2})$, a telescoping series that skips one term. The surviving terms are $\frac{1}{4}(u_1+u_2)$. We calculate $u_1 = \frac{1}{0^2+1} = 1$ and $u_2 = \frac{1}{1^2+1} = \frac{1}{2}$, so the sum is $\frac{1}{4}\left(1+\frac{1}{2}\right) = \frac{3}{8}$. ■

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Problem 4

Since $|\sin x| \le 1$ and $|\cos x| \le 1$, raising them to a high even power cannot make them larger. Thus $\sin^{2026}(x) \le \sin^2(x)$ and $\cos^{2026}(x) \le \cos^2(x)$. Adding these gives

$$\sin^{2026}(x) + \cos^{2026}(x) \le \sin^2(x) + \cos^2(x) = 1.$$

For equality we need both $\sin^{2026}(x) = \sin^2(x)$ and $\cos^{2026}(x) = \cos^2(x)$. This happens only when $\sin x \in \{0, \pm 1\}$. In the interval $[0, 2\pi]$ there are exactly 5 such points, namely $0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}, 2\pi$. ■

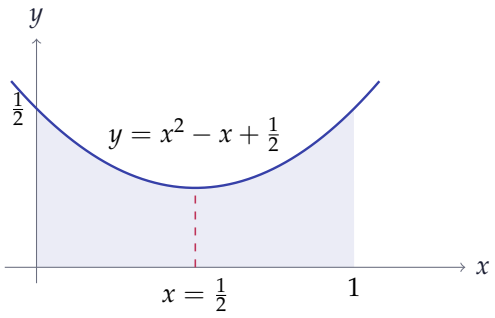
Problem 5

Integrate both sides from 0 to 1 to obtain

$$\int_0^1 f(x)\,dx + \int_0^1 f(1-x)\,dx = \int_0^1 \left(x^2 - x + \frac{1}{2}\right) dx.$$

The right side is $\left[\frac{x^3}{3} - \frac{x^2}{2} + \frac{x}{2}\right]_0^1 = \frac{1}{3} - \frac{1}{2} + \frac{1}{2} = \frac{1}{3}$. By the substitution $u = 1 - x$, the second integral on the left equals $\int_0^1 f(x)\,dx$. Thus $2\int_0^1 f(x)\,dx = \frac{1}{3}$, which gives $\int_0^1 f(x)\,dx = \frac{1}{6}$. ■

The right-hand side is symmetric about $x = \frac{1}{2}$, as the figure shows, which is consistent with the left-hand side being unchanged when x is replaced by $1 - x$.



Problem 6

The operation $a + b - ab$ comes from a factorisation, namely

$$1 - (a + b - ab) = 1 - a - b + ab = (1 - a)(1 - b).$$

Therefore the product of $1 - x$ over all numbers x on the board is invariant under every turn. Initially the product is

$$\left(1 - \frac{1}{2}\right)\left(1 - \frac{1}{3}\right)\cdots\left(1 - \frac{1}{2026}\right) = \frac{1}{2}\cdot\frac{2}{3}\cdot\frac{3}{4}\cdots\frac{2025}{2026} = \frac{1}{2026}.$$

If the final number is F, the invariant gives $1 - F = \frac{1}{2026}$, so $F = \frac{2025}{2026}$. ■

Problem 7

Put $x = 0$. Then $0 \cdot P(-1) = -3 \cdot P(0)$, so $P(0) = 0$. Since $P(0) = 0$, put $x = 1$. Then $1 \cdot P(0) = -2 \cdot P(1)$, so $P(1) = 0$. Since $P(1) = 0$, put $x = 2$. Then $2 \cdot P(1) = -1 \cdot P(2)$, so $P(2) = 0$. Thus

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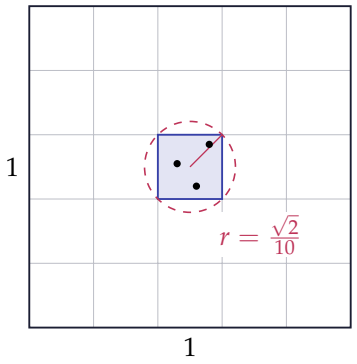
$P(x)$ has the factors x, $x-1$ and $x-2$. Let $P(x) = x(x-1)(x-2)Q(x)$ and substitute into the original equation,

$$x(x-1)(x-2)(x-3)Q(x-1) = (x-3)x(x-1)(x-2)Q(x).$$

For $x \notin \{0,1,2,3\}$ we can divide to get $Q(x-1) = Q(x)$. A periodic polynomial is constant, so $Q(x) = c$. Therefore $P(x) = cx(x-1)(x-2)$ for any real constant c. ■

Problem 8

Divide the 1×1 square into a 5×5 grid of 25 equal smaller squares, each of side length $\frac{1}{5}$. Since there are 51 points and 25 squares, by the pigeonhole principle one of these small squares contains at least $\lceil \frac{51}{25} \rceil = 3$ points.



The maximum distance between two points of one small square is the length of its diagonal, $\frac{\sqrt{2}}{5}$. Furthermore, all points of the small square are covered by its circumcircle, whose radius is half the diagonal, $r = \frac{\sqrt{2}}{10}$. We must show that this circle is no larger than a circle of radius $\frac{1}{7}$. We compare

$$\frac{\sqrt{2}}{10} \le \frac{1}{7} \iff 7\sqrt{2} \le 10 \iff 98 \le 100,$$

which is true. Thus the 3 points fit inside a disc of radius $\frac{1}{7}$. ■

Problem 9

Let $\zeta = e^{2\pi i/7}$. We know $\zeta^7 = 1$, which factors as

$$(\zeta - 1)(\zeta^6 + \zeta^5 + \zeta^4 + \zeta^3 + \zeta^2 + \zeta + 1) = 0.$$

Since $\zeta \neq 1$, the degree 6 factor is 0. Dividing by ζ^3 gives

$$(\zeta^3 + \zeta^{-3}) + (\zeta^2 + \zeta^{-2}) + (\zeta + \zeta^{-1}) + 1 = 0.$$

Let $y = \zeta + \zeta^{-1} = 2\cos(2\pi/7)$. The other powers can be written in terms of y, as $\zeta^2 + \zeta^{-2} = y^2 - 2$ and $\zeta^3 + \zeta^{-3} = y^3 - 3y$. Substituting, $(y^3 - 3y) + (y^2 - 2) + y + 1 = 0$, which gives $y^3 + y^2 - 2y - 1 = 0$. We want the roots to be $x = \cos\theta$, so we substitute $y = 2x$ to get $(2x)^3 + (2x)^2 - 2(2x) - 1 = 0$, which is

$$8x^3 + 4x^2 - 4x - 1 = 0.$$ ■

Problem 10

Subtract the second equation from the first to get $(x+y) - (y+z) = z^2 - x^2$, that is, $x - z = (z-x)(z+x)$. Rearranging gives $(x-z)(1+x+z) = 0$. By cyclic symmetry we also have $(y-x)(1+y+x) = 0$ and $(z-y)(1+z+y) = 0$.

Suppose $x = z$. Substituting back, $x + y = x^2$ and $y + x = x^2$, so $y = x^2 - x$. The third equation $x + x = y^2$ gives $2x = (x^2 - x)^2 = x^4 - 2x^3 + x^2$. This rearranges to $x^4 - 2x^3 + x^2 - 2x = 0$, which factors as $x(x-2)(x^2+1) = 0$. Thus $x = 0$ or $x = 2$, giving the triples $(0,0,0)$ and $(2,2,2)$.

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Next, suppose two variables are equal, say $x = z$, but they do not equal y. Then from $(y - x)(1 + y + x) = 0$ and $x \neq y$ we get $1 + y + x = 0$, so $y = -1 - x$. Substituting this into $y + x = x^2$ gives $-1 = x^2$, which has no real solutions.

Finally, if all three variables are distinct, then $1 + x + z = 0$ and $1 + y + x = 0$, which gives $z = y$, a contradiction. Thus the only real solutions are $(0, 0, 0)$ and $(2, 2, 2)$. ■

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